

Bayesian Inference

Philosophy, Method, and Applications

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Two Philosophies of Inference — Frequentist

Every statistical method we have discussed so far belongs to the **frequentist (or classical)** school. It rests on three postulates.

F1

Probability refers to **limiting relative frequencies**. Probabilities are objective properties of the real world.

F2

Parameters are **fixed, unknown constants**. Because they do not fluctuate, no useful probability statement can be made about them.

F3

Statistical procedures should have well-defined **long-run frequency** properties. E.g., a 95% confidence interval should trap the true parameter at least 95% of the time.

Two Philosophies of Inference — Bayesian

Bayesian inference is a competing approach, based on a different set of postulates.

B1

Probability describes **degree of belief**, not limiting frequency. We can make probability statements about anything, not just data subject to random variation.

B2

We can make probability statements about **parameters**, even though they are fixed constants.

B3

We make inferences about θ by producing a **probability distribution** for θ ; point and interval estimates are then extracted from that distribution.

Bayesian inference is controversial precisely because it embraces a subjective notion of probability. Let's set the philosophy aside and see how it is actually done.

The Bayesian Method

Bayesian inference is usually carried out in three steps.

1. Choose a probability density $f(\theta)$ — the **prior distribution** — expressing our beliefs about θ *before* seeing any data.
2. Choose a statistical model $f(x | \theta)$ reflecting our beliefs about x given θ .
3. After observing data $X_1, \dots, X_n$, update our beliefs and compute the **posterior** distribution $f(\theta | X_1, \dots, X_n)$.

The whole question is: how is step 3 actually carried out?

Bayes' Theorem

Suppose first that θ is discrete and we have one discrete observation X . Writing Θ for the parameter treated as a random variable,

$$\mathbb{P}(\Theta = \theta \mid X = x) = \frac{\mathbb{P}(X = x, \Theta = \theta)}{\mathbb{P}(X = x)} = \frac{\mathbb{P}(X = x \mid \Theta = \theta)\mathbb{P}(\Theta = \theta)}{\sum_{\theta} \mathbb{P}(X = x \mid \Theta = \theta)\mathbb{P}(\Theta = \theta)}.$$

This is **Bayes' theorem**. The version for continuous variables replaces sums and probability mass functions with integrals and densities:

Bayes' theorem — continuous case

$$f(\theta \mid x) = \frac{f(x \mid \theta)f(\theta)}{\int f(x \mid \theta)f(\theta) d\theta}.$$

From Bayes' Theorem to the Posterior

If we have n IID observations $X_1, \dots, X_n$, we replace $f(x \mid \theta)$ with

$$f(x_1, \dots, x_n \mid \theta) = \prod_{i=1}^n f(x_i \mid \theta) = \mathcal{L}_n(\theta).$$

Writing $X^n = (X_1, \dots, X_n)$ and $x^n = (x_1, \dots, x_n)$,

$$f(\theta \mid x^n) = \frac{f(x^n \mid \theta)f(\theta)}{\int f(x^n \mid \theta)f(\theta) d\theta} = \frac{\mathcal{L}_n(\theta)f(\theta)}{c_n} \propto \mathcal{L}_n(\theta)f(\theta)$$

where $c_n = \int \mathcal{L}_n(\theta)f(\theta) d\theta$ is the **normalizing constant** — it does not depend on θ .

Key relationship

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}, \quad f(\theta \mid x^n) \propto \mathcal{L}(\theta)f(\theta).$$

Throwing away c_n is harmless — it can always be recovered later if needed.

A Point Estimate — the Posterior Mean

What do we do with the posterior distribution? First, we can summarize its **center** to get a point estimate. One natural choice is the **posterior mean**:

$$\bar{\theta}_n = \int \theta f(\theta \mid x^n) d\theta = \frac{\int \theta \mathcal{L}_n(\theta) f(\theta) d\theta}{\int \mathcal{L}_n(\theta) f(\theta) d\theta}.$$

The other natural choice for the “center” of a distribution is not its mean but its **mode** — the single most plausible value of θ . That estimator is called the **MAP** estimator, and it is worth understanding on its own.

Maximum a Posteriori (MAP) Estimation

Definition — MAP estimator

The **maximum a posteriori (MAP)** estimator is the **mode** of the posterior distribution:

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} f(\theta \mid x^n).$$

Since $f(\theta \mid x^n) \propto \mathcal{L}(\theta)f(\theta)$ and maximizing a function is the same as maximizing any increasing transform of it,

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \mathcal{L}(\theta)f(\theta) = \arg \max_{\theta} \left[\underbrace{\log \mathcal{L}(\theta)}_{\ell(\theta)} + \log f(\theta) \right].$$

So the MAP estimator maximizes the **log-likelihood plus the log-prior** — the prior literally adds a term to what we are maximizing.

MAP vs. MLE — the Key Distinction

Recall the MLE maximizes the log-likelihood alone: $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta)$. Now suppose the prior is **uniform** (flat), $f(\theta) = c$ for some constant c . Then $\log f(\theta) = \log c$ is a constant that does not depend on θ , so it cannot move the arg max:

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} [\ell(\theta) + \log c] = \arg \max_{\theta} \ell(\theta) = \hat{\theta}_{\text{MLE}}.$$

The intuitive distinction

MLE is MAP under a uniform prior. Think of $\log f(\theta)$ as a landscape of bonuses and penalties laid on top of the log-likelihood. MAP tilts the likelihood surface by that landscape, pulling the maximizer toward values the prior favors. MLE assumes a perfectly flat landscape — no tilt — and lets the data alone pick the maximizer.

A Bayesian Interval Estimate

We can also obtain a Bayesian interval estimate. Find a and b such that

$$\int_{-\infty}^a f(\theta \mid x^n) d\theta = \int_b^{\infty} f(\theta \mid x^n) d\theta = \alpha/2.$$

Let $C = (a, b)$. Then

$$\mathbb{P}(\theta \in C \mid x^n) = \int_a^b f(\theta \mid x^n) d\theta = 1 - \alpha,$$

so C is a $1 - \alpha$ **posterior interval**.

Posterior Interval vs. Frequentist Confidence Interval

The two intervals are often numerically similar, but they answer **different questions**.

Frequentist confidence interval

$$\mathbb{P}_{\theta}(\theta \in C_n) = 1 - \alpha \quad \text{for every fixed } \theta.$$

θ is a fixed constant; C_n is **random** (it depends on the data). The probability is over hypothetical repetitions of the experiment, *before* the data are seen.

Bayesian posterior interval

$$\mathbb{P}(\theta \in C \mid x^n) = 1 - \alpha.$$

θ is treated as **random** (it has a posterior distribution); C is fixed once the actual data x^n are observed.

The key contrast

Frequentist: “ C_n traps θ in $1 - \alpha$ of repeated samples” — a statement about C_n , not about θ .

Bayesian: “Given the data we actually saw, θ lies in C with probability $1 - \alpha$ ” — a direct probability statement about θ itself.

Interlude — the Beta Distribution

We are about to meet the Beta distribution as a posterior, so let's recall what it is first.

Definition — Beta distribution

A random variable X has a $\text{Beta}(\alpha, \beta)$ distribution, $\alpha, \beta > 0$, if its density is

$$f(x; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1.$$

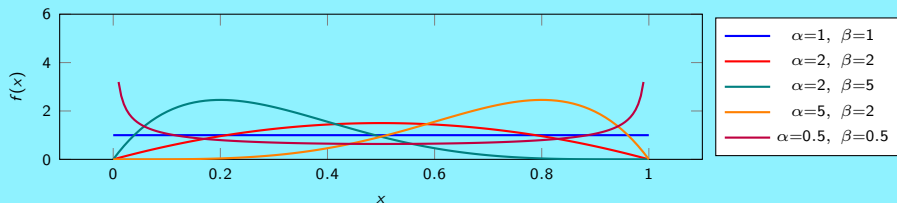
- The fraction $\Gamma(\alpha + \beta)/(\Gamma(\alpha)\Gamma(\beta))$ is **just the normalizing constant** that makes the density integrate to 1.
- Support is $[0, 1]$, so Beta is the natural family for modeling a **probability or proportion** — exactly our situation for p .
- The special case $\alpha = \beta = 1$ gives $f(x) = 1$: the $\text{Uniform}(0, 1)$ distribution.

Interlude — Mean and Shape of the Beta Distribution

Mean of a Beta distribution

$$\mathbb{E}[X] = \int_0^1 x f(x; \alpha, \beta) dx = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^\alpha (1-x)^{\beta-1} dx = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha + 1)\Gamma(\beta)}{\Gamma(\alpha + \beta + 1)} = \alpha \cdot \frac{1}{\alpha + \beta} = \boxed{\frac{\alpha}{\alpha + \beta}}$$

(using $\Gamma(z + 1) = z\Gamma(z)$, since the middle integral is the normalizing constant of a $\text{Beta}(\alpha + 1, \beta)$ density).



Example — Bernoulli with a Uniform Prior: Setup

Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$ and take the uniform prior $f(p) = 1$. By Bayes' theorem, the posterior satisfies

$$f(p \mid x^n) \propto f(p) \mathcal{L}_n(p) = 1 \cdot p^s (1 - p)^{n-s}, \quad s = \sum_{i=1}^n x_i.$$

Rewrite the exponents to match a familiar shape:

$$f(p \mid x^n) \propto p^{(s+1)-1} (1 - p)^{(n-s+1)-1}.$$

Example — Bernoulli with a Uniform Prior: Recognizing Beta

Compare this to the $\text{Beta}(\alpha, \beta)$ density, which is proportional to $p^{\alpha-1}(1-p)^{\beta-1}$.
Matching exponents:

$$\alpha - 1 = s \implies \alpha = s + 1, \quad \beta - 1 = n - s \implies \beta = n - s + 1.$$

Conclusion

$$p \mid x^n \sim \text{Beta}(s + 1, n - s + 1).$$

Notice we found the normalizing constant $\Gamma(n+2)/(\Gamma(s+1)\Gamma(n-s+1))$ **without ever doing the integral** — just by pattern-matching.

Example — Bernoulli: the Bayes Estimator

Using the Beta-mean formula $\alpha/(\alpha + \beta)$ from before, the posterior mean (the Bayes estimator) is

$$\bar{p} = \frac{s + 1}{n + 2}.$$

It is instructive to rewrite this as

$$\bar{p} = \lambda_n \hat{p} + (1 - \lambda_n) \tilde{p}$$

where $\hat{p} = s/n$ is the MLE, $\tilde{p} = 1/2$ is the prior mean, and $\lambda_n = n/(n + 2) \approx 1$ for large n . The Bayes estimator is a **weighted average of the data and the prior**, with the data dominating as n grows.

Example — Bernoulli: the MAP Estimator

For $a, b > 1$, the mode of $\text{Beta}(a, b)$ is found by differentiating $x^{a-1}(1-x)^{b-1}$ and setting the result to 0:

$$(a-1)x^{a-2}(1-x)^{b-1} = (b-1)x^{a-1}(1-x)^{b-2} \implies (a-1)(1-x) = (b-1)x \implies x = \frac{a-1}{a+b-2}.$$

Here $a = s + 1$, $b = n - s + 1$, so

$$\hat{p}_{\text{MAP}} = \frac{(s+1) - 1}{(s+1) + (n-s+1) - 2} = \frac{s}{n} = \hat{p}_{\text{MLE}}.$$

Confirms the intuition

Because the prior was **uniform**, MAP collapses exactly to the MLE. Under a general $\text{Beta}(\alpha, \beta)$ prior (next slide), $\hat{p}_{\text{MAP}} = (\alpha + s - 1)/(\alpha + \beta + n - 2) \neq \hat{p}_{\text{MLE}}$ in general — it only reduces to the MLE when $\alpha = \beta = 1$.

Example — A General Beta Prior

Suppose instead we use the prior $p \sim \text{Beta}(\alpha, \beta)$. Repeating the same calculation,

$$f(p \mid x^n) \propto p^{\alpha-1}(1-p)^{\beta-1} \cdot p^s(1-p)^{n-s} = p^{(\alpha+s)-1}(1-p)^{(\beta+n-s)-1},$$

so $p \mid x^n \sim \text{Beta}(\alpha + s, \beta + n - s)$. The flat prior is just the special case $\alpha = \beta = 1$.

The posterior mean is

$$\bar{p} = \frac{\alpha + s}{\alpha + \beta + n} = \left(\frac{n}{\alpha + \beta + n} \right) \hat{p} + \left(\frac{\alpha + \beta}{\alpha + \beta + n} \right) p_0, \quad p_0 = \frac{\alpha}{\alpha + \beta}.$$

Definition

When the prior and the posterior belong to the **same family** of distributions, we say the prior is **conjugate** with respect to the model.

In the Bernoulli example, the Beta distribution is conjugate for the Bernoulli (and Binomial) model: a Beta prior always produces a Beta posterior.

Example — Normal with a Normal Prior: Result

Let $X_1, \dots, X_n \sim N(\theta, \sigma^2)$ with σ known, and take the prior $\theta \sim N(a, b^2)$. The posterior turns out to be

$$\theta \mid X^n \sim N(\bar{\theta}, \tau^2)$$

where

$$\bar{\theta} = w\bar{X} + (1 - w)a, \quad w = \frac{1/\text{se}^2}{1/\text{se}^2 + 1/b^2}, \quad \frac{1}{\tau^2} = \frac{1}{\text{se}^2} + \frac{1}{b^2},$$

and $\text{se} = \sigma/\sqrt{n}$ is the standard error of the MLE $\bar{X}$. This is another example of a **conjugate** prior.

As $n \rightarrow \infty$, $w \rightarrow 1$ and $\tau/\text{se} \rightarrow 1$, so the posterior becomes $N(\hat{\theta}, \text{se}^2)$ — the same happens if $b \rightarrow \infty$, i.e. the prior becomes very flat.

Example — Normal: the Posterior Interval

Let us find $C = (c, d)$ with $\mathbb{P}(\theta < c \mid X^n) = \mathbb{P}(\theta > d \mid X^n) = .025$. Standardizing,

$$\begin{aligned}\mathbb{P}(\theta < c \mid X^n) &= \mathbb{P}\left(\frac{\theta - \bar{\theta}}{\tau} < \frac{c - \bar{\theta}}{\tau} \mid X^n\right) \\ &= \mathbb{P}\left(Z < \frac{c - \bar{\theta}}{\tau}\right) = .025.\end{aligned}$$

Since $\mathbb{P}(Z < -1.96) = .025$, we need $(c - \bar{\theta})/\tau = -1.96$, so $c = \bar{\theta} - 1.96\tau$. By the same argument, $d = \bar{\theta} + 1.96\tau$.

95% Bayesian interval

$$\bar{\theta} \pm 1.96 \tau.$$

Since $\bar{\theta} \approx \hat{\theta}$ and $\tau \approx \text{se}$ for large n , this matches the **frequentist** confidence interval $\hat{\theta} \pm 1.96 \text{se}$.

How do we make inferences about a function $\tau = g(\theta)$? We apply the same reasoning used to find the density of a transformed random variable.

Posterior CDF method

$$H(\tau | x^n) = \mathbb{P}(g(\theta) \leq \tau | x^n) = \int_A f(\theta | x^n) d\theta, \quad A = \{\theta : g(\theta) \leq \tau\}.$$

The posterior density is then $h(\tau | x^n) = H'(\tau | x^n)$.

Example — the Log-Odds Transform: Setup

Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$ with $f(p) = 1$, so $p \mid X^n \sim \text{Beta}(s + 1, n - s + 1)$. Let

$$\psi = \log\left(\frac{p}{1 - p}\right)$$

be the log-odds (*logit*) of p . We want the posterior density $h(\psi \mid x^n)$.

Example — the Log-Odds Transform: Derivation

$$\begin{aligned} H(\psi \mid x^n) &= \mathbb{P}(\Psi \leq \psi \mid x^n) = \mathbb{P}\left(\log \frac{P}{1-P} \leq \psi \mid x^n\right) = \mathbb{P}\left(P \leq \frac{e^\psi}{1+e^\psi} \mid x^n\right) \\ &= \int_0^{e^\psi/(1+e^\psi)} f(p \mid x^n) dp = \frac{\Gamma(n+2)}{\Gamma(s+1)\Gamma(n-s+1)} \int_0^{e^\psi/(1+e^\psi)} p^s (1-p)^{n-s} dp. \end{aligned}$$

Differentiating with respect to ψ (chain rule, with $\partial/\partial\psi [e^\psi/(1+e^\psi)] = e^\psi/(1+e^\psi)^2$):

$$\begin{aligned} h(\psi \mid x^n) &= H'(\psi \mid x^n) \\ &= \frac{\Gamma(n+2)}{\Gamma(s+1)\Gamma(n-s+1)} \left(\frac{e^\psi}{1+e^\psi}\right)^s \left(\frac{1}{1+e^\psi}\right)^{n-s} \cdot \frac{e^\psi}{(1+e^\psi)^2} \\ &= \frac{\Gamma(n+2)}{\Gamma(s+1)\Gamma(n-s+1)} \left(\frac{e^\psi}{1+e^\psi}\right)^s \left(\frac{1}{1+e^\psi}\right)^{n-s+2}, \quad \psi \in \mathbb{R}. \end{aligned}$$

Approximating the Posterior by Simulation

The posterior can often be approximated by simulation. Draw $\theta_1, \dots, \theta_B \sim p(\theta \mid x^n)$. Then:

- A histogram of $\theta_1, \dots, \theta_B$ approximates $p(\theta \mid x^n)$.
- The posterior mean is approximated by $B^{-1} \sum_{j=1}^B \theta_j$.
- The $1 - \alpha$ posterior interval is approximated by $(\theta_{\alpha/2}, \theta_{1-\alpha/2})$, the sample quantiles.
- If $\tau_i = g(\theta_i)$, then $\tau_1, \dots, \tau_B$ is a sample from $f(\tau \mid x^n)$ — **no analytic calculus required**.

Example — Simulating the Log-Odds Posterior

We can approximate the posterior for ψ from the earlier example without doing any calculus at all:

1. Draw $P_1, \dots, P_B \sim \text{Beta}(s + 1, n - s + 1)$.
2. Let $\psi_i = \log(P_i / (1 - P_i))$ for $i = 1, \dots, B$.

Now $\psi_1, \dots, \psi_B$ are IID draws from $h(\psi \mid x^n)$ — a histogram of these values estimates $h(\psi \mid x^n)$, matching the formula we derived by hand.

Where Does the Prior Come From?

One school of thought, **subjectivism**, says the prior should reflect our genuine subjective belief about θ before seeing data. This can be impractical with many parameters, and seems to inject subjectivity into otherwise objective science.

An alternative is to seek a **noninformative prior** that lets the data speak for itself. The obvious candidate is a **flat prior** $f(\theta) \propto \text{constant}$.

Improper Priors

Let $X \sim N(\theta, \sigma^2)$ with σ known, and adopt the flat prior $f(\theta) \propto c$ for a constant $c > 0$. Since $\int f(\theta) d\theta = \infty$, this is not a genuine probability density — we call it an **improper prior**.

We can still formally apply Bayes' theorem:

$$f(\theta | x^n) \propto \mathcal{L}_n(\theta)f(\theta) \propto \mathcal{L}_n(\theta).$$

This gives $\theta | X^n \sim N(\bar{X}, \sigma^2/n)$, matching the frequentist point and interval estimators exactly.

Rule of thumb

Improper priors are not a problem as long as the resulting **posterior** is a well-defined probability distribution.

Flat Priors Are Not Invariant

Let $X \sim \text{Bernoulli}(p)$ and use the flat prior $f(p) = 1$, representing ignorance about p . Let $\psi = \log(p/(1-p))$, so $p = e^\psi / (1 + e^\psi)$ and $dp/d\psi = e^\psi / (1 + e^\psi)^2$. By the change-of-variables formula,

$$f_\psi(\psi) = f_P(p) \left| \frac{dp}{d\psi} \right| = \frac{e^\psi}{(1 + e^\psi)^2},$$

which is **not flat**.

The paradox

If we are ignorant about p , we should also be ignorant about $\psi = \log(p/(1-p))$ — but a flat prior on p does *not* give a flat prior on ψ . Flat priors are not **transformation invariant**.

Jeffreys came up with a rule for constructing priors that resolves the paradox above.

Jeffreys' rule

$$f(\theta) \propto I(\theta)^{1/2}$$

where $I(\theta)$ is the **Fisher information function**.

This rule turns out to be **transformation invariant** — it gives the same prior no matter how we parametrize the model. But what exactly is $I(\theta)$?

Interlude — What Is Fisher Information?

Definition — score function

For a model $f(x; \theta)$, the **score function** is $s(\theta; x) = \frac{\partial}{\partial \theta} \log f(x; \theta)$.

Definition — Fisher information

$$I(\theta) = \mathbb{E}_{\theta} [s(\theta; X)^2] = \mathbb{E}_{\theta} \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \right] = -\mathbb{E}_{\theta} \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right]$$

(the last equality holds under mild regularity conditions).

Intuition: $I(\theta)$ measures how sharply **curved** the log-likelihood is near θ — how much information one observation carries about the parameter.

Interlude — Fisher Information for the Bernoulli

For $f(x; p) = p^x(1 - p)^{1-x}$, $\log f(x; p) = x \log p + (1 - x) \log(1 - p)$.

Score:

$$\frac{\partial}{\partial p} \log f = \frac{x}{p} - \frac{1 - x}{1 - p}.$$

Second derivative:

$$\frac{\partial^2}{\partial p^2} \log f = -\frac{x}{p^2} - \frac{1 - x}{(1 - p)^2}.$$

Take $-\mathbb{E}$, using $\mathbb{E}[X] = p$ and $\mathbb{E}[1 - X] = 1 - p$:

$$I(p) = \frac{p}{p^2} + \frac{1 - p}{(1 - p)^2} = \frac{1}{p} + \frac{1}{1 - p}.$$

Result

$$I(p) = \frac{1}{p(1 - p)}.$$

Example — Jeffreys' Prior for the Bernoulli

Jeffreys' rule says to use

$$f(p) \propto \sqrt{I(p)} = p^{-1/2}(1-p)^{-1/2}.$$

This is a familiar shape

This is exactly the $\text{Beta}(1/2, 1/2)$ density — the U-shaped purple curve from our very first Beta plot! It is very close to a uniform density, but places slightly more mass near 0 and 1.

Multiparameter Problems

Suppose $\theta = (\theta_1, \dots, \theta_p)$. The posterior density is still

$$f(\theta \mid x^n) \propto \mathcal{L}_n(\theta)f(\theta).$$

To make inferences about a single θ_1 , we need its **marginal posterior**:

$$f(\theta_1 \mid x^n) = \int \cdots \int f(\theta_1, \dots, \theta_p \mid x^n) d\theta_2 \cdots d\theta_p,$$

which may not be feasible to compute analytically.

Simulation avoids the integral entirely. Draw vectors

$$\theta^1, \dots, \theta^B \sim f(\theta \mid x^n),$$

where the superscript indexes the draw, $\theta^j = (\theta_1^j, \dots, \theta_p^j)$.

Collecting just the first coordinate of every draw,

$$\theta_1^1, \dots, \theta_1^B,$$

gives a sample from $f(\theta_1 \mid x^n)$ — we have avoided doing any integrals.

Example — Comparing Two Binomials: Setup

Suppose we have n_1 control patients and n_2 treatment patients, with X_1 control and X_2 treatment patients surviving. We want to estimate $\tau = g(p_1, p_2) = p_2 - p_1$, where

$$X_1 \sim \text{Binomial}(n_1, p_1), \quad X_2 \sim \text{Binomial}(n_2, p_2).$$

With the flat prior $f(p_1, p_2) = 1$, the posterior is

$$f(p_1, p_2 \mid x_1, x_2) \propto p_1^{x_1} (1 - p_1)^{n_1 - x_1} p_2^{x_2} (1 - p_2)^{n_2 - x_2}.$$

Example — Comparing Two Binomials: Independence

The posterior **factorizes**:

$$f(p_1, p_2 \mid x_1, x_2) = \underbrace{[p_1^{x_1}(1-p_1)^{n_1-x_1}]}_{\propto f(p_1|x_1)} \cdot \underbrace{[p_2^{x_2}(1-p_2)^{n_2-x_2}]}_{\propto f(p_2|x_2)} = f(p_1 \mid x_1)f(p_2 \mid x_2).$$

Conclusion

p_1 and p_2 are **independent** under the posterior, with

$$p_1 \mid x_1 \sim \text{Beta}(x_1 + 1, n_1 - x_1 + 1), \quad p_2 \mid x_2 \sim \text{Beta}(x_2 + 1, n_2 - x_2 + 1).$$

Example — Comparing Two Binomials: Simulating τ

Draw independently

$$P_{1,1}, \dots, P_{1,B} \sim \text{Beta}(x_1 + 1, n_1 - x_1 + 1), \quad P_{2,1}, \dots, P_{2,B} \sim \text{Beta}(x_2 + 1, n_2 - x_2 + 1).$$

Then

$$\tau_b = P_{2,b} - P_{1,b}, \quad b = 1, \dots, B,$$

is a sample from $f(\tau \mid x_1, x_2)$ — again, no integrals needed.

Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ for scalar θ . The Bayesian approach puts a prior on H_0 and on θ , then computes $\mathbb{P}(H_0 \mid X^n)$.

It is usually reasonable to use $\mathbb{P}(H_0) = \mathbb{P}(H_1) = 1/2$ (though not essential). Under H_1 we need a prior density $f(\theta)$ for θ .

From Bayes' theorem,

$$\begin{aligned}\mathbb{P}(H_0 \mid X^n = x^n) &= \frac{f(x^n \mid H_0)\mathbb{P}(H_0)}{f(x^n \mid H_0)\mathbb{P}(H_0) + f(x^n \mid H_1)\mathbb{P}(H_1)} \\ &= \frac{\frac{1}{2}f(x^n \mid \theta_0)}{\frac{1}{2}f(x^n \mid \theta_0) + \frac{1}{2} \int f(x^n \mid \theta)f(\theta) d\theta} \\ &= \frac{f(x^n \mid \theta_0)}{f(x^n \mid \theta_0) + \int f(x^n \mid \theta)f(\theta) d\theta} = \frac{\mathcal{L}(\theta_0)}{\mathcal{L}(\theta_0) + \int \mathcal{L}(\theta)f(\theta) d\theta}.\end{aligned}$$

Bayesian Testing — a Prior-Free Bound

Unlike in estimation, we **cannot use improper priors** for testing — that would leave an undefined constant in the denominator. So the prior $f(\theta)$ must be chosen carefully.

Since the likelihood is maximized at the MLE $\hat{\theta}$,

$$0 \leq \int \mathcal{L}(\theta) f(\theta) d\theta \leq \mathcal{L}(\hat{\theta}).$$

This gives the bound

$$\frac{\mathcal{L}(\theta_0)}{\mathcal{L}(\theta_0) + \mathcal{L}(\hat{\theta})} \leq \mathbb{P}(H_0 \mid X^n = x^n) \leq 1.$$

The upper bound is trivial, but the **lower bound is non-trivial and prior-free**.

Strengths and Weaknesses of Bayesian Inference

Bayesian inference is appealing when prior information is available, since Bayes' theorem is a natural way to combine it with data. It is also **psychologically** appealing: it lets us make direct probability statements about θ .

Frequentist confidence sets C_n trap θ with limiting frequency 95% — but we cannot say $\mathbb{P}(\theta \in C_n \mid X^n) = .95$. The probability statement is about C_n , not θ .

In parametric models with large samples, Bayesian and frequentist inference tend to **agree**. In general, they need not.

Example — a Confidence Set Puzzle: Setup

Let θ be a fixed, known real number, and let X_1, X_2 be independent with $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = 1/2$. Define $Y_i = \theta + X_i$, and suppose we observe only Y_1 and Y_2 . Let

$$C = \begin{cases} \{Y_1 - 1\} & \text{if } Y_1 = Y_2 \\ \{(Y_1 + Y_2)/2\} & \text{if } Y_1 \neq Y_2. \end{cases}$$

No matter what θ is, $\mathbb{P}_\theta(\theta \in C) = 3/4$, so C is a valid **75% confidence set**.

Example — a Confidence Set Puzzle: the Puzzle

Suppose we observe $Y_1 = 15$, $Y_2 = 17$. Since $Y_1 \neq Y_2$,

$$C = \{(15 + 17)/2\} = \{16\}.$$

The discomfort

Because $Y_1 \neq Y_2$, exactly one X_i equals $+1$ and the other -1 , so $\theta = (Y_1 + Y_2)/2$ **exactly** — we are certain that $\theta = 16$. Yet by construction, all we are entitled to call this is a “75% confidence set.” This bothers many people.

Example — a Confidence Set Puzzle: the Bayesian Resolution

Assume θ is an integer with prior $f(\theta) > 0$ for every integer. When $Y = (15, 17)$, the likelihood is

$$\mathcal{L}(\theta) = \begin{cases} 1/4 & \theta = 16 \\ 0 & \text{otherwise.} \end{cases}$$

Applying Bayes' theorem,

$$\mathbb{P}(\Theta = \theta \mid Y = (15, 17)) = \begin{cases} 1 & \theta = 16 \\ 0 & \text{otherwise.} \end{cases}$$

Resolution

Hence $\mathbb{P}(\theta \in C \mid Y = (15, 17)) = 1$ — the Bayesian posterior makes exactly the probability statement about θ that our intuition wanted, with no contradiction.

Key Takeaway — When Not to Use Bayesian Inference

Be cautious when...

- The problem is **high-dimensional or nonparametric**. Bayesian methods are slaves to the likelihood function; if the likelihood is badly behaved, Bayesian inference can fail badly even while simple frequentist alternatives work fine.
- You need **Bayesian testing**. Improper priors cannot be used, and results can be highly sensitive to the choice of prior — this requires great care.
- You need **guaranteed long-run performance** (e.g. valid coverage) no matter what the truth is — that is a frequentist guarantee, not a Bayesian one.

Use Bayesian inference to combine **genuine prior information** with data in a well-specified, lower-dimensional model. Use frequentist methods when you need procedures with guaranteed long-run behavior.